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PS ID : EC602

Solution Title : KisanCall — Voice AI for Farmersl

Team Name : EL Bicho

College Name : Thapar University

Team Leader : Kunal Singla

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KisanCall — Voice AI for Farmers

- Farmers lack access to real-time expert advice
- Language barrier (most tools are English-based)
- No easy way to ask questions via voice

Farmers speak in **Punjabi/Hindi**
System understands and responds instantly
Covers:
Crop disease
Weather
Mandi prices
Govt schemes

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Pipeline

4 SLM Architecture

STT — Whisper small (244M params)
Guardrails — Phi-3 mini 3.8B
Intent Router — Phi-3 mini 3.8B:
Response Generator — Phi-3 mini 3.8B

Why Small Language Models?

Faster
Lower cost
Works on small devices
Better for real-time systems

Speech-to-Text (STT)

Based on Whisper Small (244M params)
Fine-tuned for Punjabi + Hindi
Handles agricultural vocabulary

Data Preparation

Used Punjabi + Hindi speech datasets
Added synthetic farming phrases using TTS
Improved domain understanding

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Fine-tuning

Used LoRA (efficient fine-tuning)
Only ~3% parameters trained
Faster + low memory

Real-time Inference

Voice Activity Detection (detect speech)
Converts speech → text
~120ms latency

Guardrails SLM

Checks if query is farming-related
Blocks unsafe content
Reduces unnecessary computation

Future Modules

Intent Router → classify query
Response Generator → answer
TTS → convert to voice

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Guardrails SLM

Binary classifier (farming vs non-farming)

Filters unsafe queries

Runs in <50ms

Intent Router SLM

Crop disease

Weather

Mandi prices

Govt schemes

Response Generator SLM

Generates final answer

Uses agricultural knowledge

Keeps responses short and voice-friendly

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Why This Architecture is Powerful

Faster than one big model

Modular (easy to improve)

Better domain accuracy

Lower cost

Works in real-time

Technical Innovation

Fine-tuned Whisper for Punjabi/Hindi

Domain-specific dataset (agriculture)

LoRA for efficient training

Modular SLM pipeline

Real-time inference (~120ms)

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Future Extensions

Add image input (crop photo)
Integrate live mandi APIs
Offline mobile version
Regional language expansion

Why not use GPT??

Too slow
Too expensive
Not domain-specific

How KisanCall Benefits Farmers

Instant Crop Disease Help

Describe symptoms in Punjabi → get treatment advice in seconds, no visit needed.

Live Mandi Price Check

Real-time wheat, paddy & mustard prices so farmers sell at the right moment.

Hyperlocal Weather Alerts

Rain & frost warnings by district — plan sowing, harvesting and spraying wisely.

Government Scheme Eligibility

PM Kisan, crop insurance, KCC — explained in their own language, instantly.

Under 500ms Response

Faster than a Google search. No waiting — a real-time voice conversation.

Works on ANY Phone

No smartphone needed. Call a toll-free number and speak naturally — like a friend.

58%	2	<500ms	₹0
of India's workforce in agriculture	languages supported (Punjabi & Hindi)	end-to-end pipeline latency	cost to the farmer

KisanCall — Growth Roadmap & Business Vision

01

Foundation

LIVE

- Whisper STT fine-tuned on Punjabi + Hindi agri vocab
- Phi-3 mini Guardrails SLM via Oumi (LoRA)
- <500ms end-to-end · WER 16% Punjabi / 9% Hindi
- Hackathon demo — fully functional pipeline

02

Intelligence

30 DAYS

- Intent Router SLM (crop / mandi / weather / schemes)
- RAG over ICAR crop disease database
- Agmarknet API — live mandi prices
- Response Generator fine-tuned on KVK docs

03

Reach

90 DAYS

- IVR integration — toll-free number for basic phones
- 50,000+ concurrent calls supported
- Automated alerts: pest warnings, rain forecasts
- WhatsApp Business API integration

04

Scale

VISION

- Offline edge deployment on ₹3,000 Android devices
- 10+ Indian regional languages
- Personalised crop calendar per farmer profile
- State govt portal integrations & B2G revenue

"Every phase, one goal — putting expert agricultural knowledge in every farmer's pocket."